Regime-Dependent Predictive Structure Between Equity Factors: Evidence from Granger Causality

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Abstract

We evaluate regime-dependent predictive structure between equity factors using 35 years of Fama-French daily data (1990–2024, 8,817 trading days) and a Student-t HMM with 50 multi-start initializations. To eliminate regime-identification circularity, the *primary* analysis uses a frozen out-of-sample (OOS) design: HMM trained on 1990–2012 (n = 5,797 days), applied without refitting to 2013–2024 (n = 3,020 days).

The central finding is a post-GFC regime shift: HML→SMB is strongly significant in the Normal regime (full-sample $p = 8.75 \times 10^{-9}$, Bonferroni-significant) but *pre-GFC only* (Chow $p = 2.29 \times 10^{-6}$; post-2008: $p = 0.73$). In the frozen OOS evaluation, HML→SMB shows significant evidence in the Elevated regime: F-$p = 0.014$, HAC $p = 0.041$, permutation $p = 0.022$ (50,000 shuffles, SE = 0.0007, 95% CI $[0.021, 0.023]$, $n = 953$), $\Delta R^2 = 0.73\%$. (Economic magnitude is modest: $\Delta R^2 < 1\%$ throughout, so statistical significance does not imply economic importance.) Normal and Crisis null. Two of 3 local optima (41/50 seeds) show this pattern. Trivariate robustness (MKT-RF control) slightly strengthens the Elevated result (F-$p = 0.012$, HAC $p = 0.035$); MKT-RF is non-significant in Elevated and Normal (F-$p \geq 0.43$), marginally so in Crisis (F-$p = 0.077$). A four-model diagnostic (Linear, RF, MLP, LSTM) finds no nonlinear improvement over the linear baseline; transfer entropy identifies an additional reverse nonlinear channel (SMB→HML, empirical $p < 0.005$, sensitivity fit, seed 42) in Normal. The effect does not translate to profitable trading (Sharpe = −0.07), but a hybrid VaR detector achieves 5.60% violations (CC $p = 0.336$). We treat Granger findings as predictive precedence, not structural causality.

CCS Concepts

• **Mathematics of computing → Time series analysis**; • **Computing methodologies → Causal reasoning and diagnostics**.

Keywords

Factor Investing, Granger Causality, Regime Switching, Hidden Markov Models, Model Complexity Characterization, Out-of-Sample Validation, Risk Management

ACM Reference Format:

Anonymous Author(s). 2026. Regime-Dependent Predictive Structure Between Equity Factors: Evidence from Granger Causality. In *Proceedings of* . ACM, New York, NY, USA, 17 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

The August 2007 quantitative meltdown, in which systematic equity strategies lost more than 27% in three trading days (August 7–9, 2007) [60], revealed a critical blind spot in factor-based risk management. When multiple quantitative funds held similar factor exposures, forced liquidation by one fund created price pressure that cascaded to all others. Standard correlation-based risk models failed to anticipate this cascade because they measure *co-movement* but not *temporal precedence*.

The Missing Piece: Which Factor Moves First?

Existing research establishes three stylized facts: (1) Equity return correlations increase during market stress [6]; (2) Returns exhibit regime-switching behavior [50]; (3) Factor crowding amplifies drawdowns during liquidation [81].

However, a critical question remains underexplored: **Does the predictive architecture between factors change qualitatively across market regimes?**

Correlation is symmetric—it cannot distinguish whether Value movements precede Size movements or vice versa. Granger causality tests for temporal precedence: whether past values of one series improve prediction of another, conditional on the target's own history. If such predictive relationships vary by regime, we could:

- Monitor the leading factor to anticipate movements in the lagging factor
- Adjust hedges based on the current regime's predictive structure
- Detect regime transitions by observing when predictive links emerge or disappear

Important caveat. Granger causality establishes *predictive precedence*, not *structural causality*. A statistically significant Granger relationship could arise from: (1) direct causal influence, (2) a common driver affecting both series with different lags, or (3) omitted variables. We interpret our findings as documenting regime-dependent predictive structure, with economic mechanisms offered as hypotheses rather than verified causal channels—a distinction foundational in the causal inference literature [72] and articulated for factor research by López de Prado [64]. We adopt the Granger framework for its tractability on high-dimensional daily factor series. López de Prado advocates structural causal models (SCMs/DAGs) as the more rigorous alternative for factor research; we acknowledge this distinction throughout.

1.1 Our Findings

Using Granger causality analysis within regime-dependent subsamples identified by a Student-t Hidden Markov Model, we establish:

Main Finding: Bonferroni-significant in-sample structure with significant OOS evidence. The in-sample Normal-regime result is unambiguously significant ($p = 8.75 \times 10^{-9}$, Bonferroni-significant across 30 pairs) with a confirmed structural break at January 2008 (Chow $p = 2.29 \times 10^{-6}$). Using a frozen OOS design (HMM trained on 1990–2012, Granger tested on 2013–2024), HML→SMB shows suggestive evidence in the Elevated regime with no circularity (economically motivated pair; ranks 2nd of 30 directed pairs by F-statistic; see Section 3.5):

- *Elevated (OOS, $n = 953$):* $p = 0.014$, HAC $p = 0.041$ (Newey–West, bandwidth 1), $\Delta R^2 = 0.73\%$ (lag 1, pre-fixed from in-sample BIC); permutation $p = 0.022$ (50,000 label shuffles [43], SE = 0.0007, 95% CI [0.021, 0.023])—significant and consistent with the in-sample pattern.

- *Normal (OOS):* $p = 0.149$ (null).
- *Crisis (OOS):* $p = 0.281$ (null).

A full 50-seed sensitivity analysis finds that 2 of 3 local optima show Elevated significance (F-$p < 0.05$), covering 41 of 50 seeds—these represent 2 distinct HMM solutions, not 41 independent ones. The 9 null seeds all share the lowest-LL local optimum. In-sample analysis (1990–2024) provides higher-power corroboration: Normal-regime $p = 8.75 \times 10^{-9}$ (HAC $p = 9.45 \times 10^{-8}$), surviving Bonferroni correction across 30 directed pairs. A single-stage Markov-Switching VAR further supports HML lags ($p < 10^{-12}$), and event tests are mixed (2/6 marginally significant at $p < 0.10$, 4/6 in the predicted direction) (Section 5.1).

Predominantly linear. A four-model diagnostic protocol (Linear, RF, MLP, LSTM) finds that no nonlinear method improves HML→SMB prediction beyond the linear baseline. However, transfer entropy reveals an additional nonlinear reverse channel (SMB→HML, $p < 0.005$ empirical; sensitivity fit, seed 42; non-significant under primary fit; $z = 5.37$ under Gaussian approximation, unreliable for heavy-tailed data) in Normal, invisible to standard Granger tests (Section 5.6).

Mechanism evidence (sensitivity fit, seed 42). Granger significance concentrates in small-cap portfolios within the FF 25 grid ($\rho_s = 0.35$, permutation $p = 0.046$), consistent with portfolio overlap; this analysis uses an alternative HMM fit optimized for crisis detection rather than the primary OOS fit (see Section 4.5).

Practical Implication. The effect is small ($\Delta R^2 \approx 1\%$) and does not translate to profitable trading, but a hybrid VaR model combining HMM regime detection with a realized-volatility fallback achieves 5.60% violation rate ($p_{\text{CC}} = 0.336$)—the best among all specifications (Section 5.5).

1.2 Contributions

(1) **Structural regime shift (Bonferroni-significant):** HML→SMB is strongly significant in the Normal regime over the full sample ($p = 8.75 \times 10^{-9}$, Bonferroni-significant, $\Delta R^2 = 2.06\%$ pre-GFC) but undergoes a structural break at January 2008 (Chow $p = 2.29 \times 10^{-6}$; Bai-Perron sup-$F = 22.1$; post-2008 Normal: $p = 0.73$). The regime-conditional architecture—present before the GFC, absent after—is the paper's central empirical finding (Section 4.3).

(2) **Frozen OOS evidence:** Consistent with the pre-GFC Normal finding, HML→SMB shows significant evidence in the Elevated regime of the frozen OOS window (2013–2024, $n = 953$): F-$p = 0.014$, HAC $p = 0.041$, permutation $p = 0.022$ (50,000 shuffles, SE = 0.0007, 95% CI [0.021, 0.023]); HML→SMB ranks 2nd of 30 directed pairs in the OOS Elevated window (rank-based max-statistic $p = 0.067$); 41/50 seeds show significance. HAC bandwidth sensitivity (1, 4, 7, 10, 15) yields $p < 0.05$ throughout (range: 0.015–0.025), confirming robustness (Sections 5.2, 3.5).

(3) **Methodological:** Student-t HMM with multi-start selection (50 initializations) for reproducible regime detection, plus a four-model complexity diagnostic (Linear, RF, MLP, LSTM) establishing the link as predominantly linear (Sections 3.2, 4.2, 5.6).

(4) **Exploratory mechanism and risk-monitoring evidence:** Portfolio overlap analysis (FF 25; sensitivity fit, seed 42; $\rho_s = 0.35$, permutation $p = 0.046$) and a hybrid VaR detector (5.60% violations, CC $p = 0.336$) provide exploratory mechanism evidence and a practical application pathway (Sections 4.5, 5.4).

2 Related Work

Factor models and cross-sectional returns. Fama and French [36, 37] establish that Size, Value, Profitability, and Investment factors explain cross-sectional return variation. Asness et al. [9] document value and momentum effects globally. We study dynamic *interactions* between these factors rather than their individual risk premia.

Factor time-series predictability. Jegadeesh and Titman [58] establish cross-sectional momentum (past winners outperform past losers), with Carhart [21] formalizing MOM as a systematic factor. Ehsani and Linnainmaa [34] show that factor returns exhibit time-series momentum: past factor returns predict future factor returns, with the momentum factor loading on lagged returns across all other factors. Gupta and Kelly [48] extend this finding across asset classes. Haddad, Kozak and Santosh [49] show that individual factor returns are predictable from conditioning variables representing the stochastic discount factor (the most predictable SDF components achieve out-of-sample $R^2 \approx 4\%$ at the monthly frequency)—a result on factor-level predictability from SDF-motivated conditioning variables. Flögel, Schlag and Zunft [40] exploit within-factor (own) first-order autocorrelations in *monthly* factor returns to construct momentum-managed factor portfolios. Barroso and Santa-Clara [12] and Daniel and Moskowitz [31] document that momentum profits are regime-dependent: high in tranquil periods, negative during momentum-crash episodes. Neuhierl et al. [68] show that past factor returns predict future factor returns across 300+ factors, with value and profitability timing improving most—a general cross-factor predictability result. **Gap:** These papers establish within-factor time-series momentum, regime-dependent momentum *returns*, SDF-based factor timing, cross-factor autocorrelation-based portfolio management (Flögel et al.), and general cross-factor predictability (Neuhierl et al.), but do not examine whether the *Granger predictive structure between distinct factors*—which factor leads the other, and under which regime—changes qualitatively across market states. Our contribution is distinct from factor momentum by econometric construction: the Granger test conditions on SMB's own lags, ruling out within-factor SMB autocorrelation as an explanation; the HML coefficient tests *incremental* cross-factor content beyond SMB's history, and our regime conditioning reveals Normal-regime significance invisible to the unconditional autocorrelation approach of Flögel et al. This design reveals regime-dependence that unconditional tests miss: significant at $p = 8.75 \times 10^{-9}$ in Normal (Bonferroni-significant) but null in Crisis ($p = 0.695$; the Crisis regime contains only 1,071 days, limiting power relative to Normal's 4,723 days). Additionally, Chordia and Shivakumar [26] document that momentum profits vary systematically with the business cycle; Cooper, Gutierrez and Hameed [29] show momentum profits are strongly state-conditional (positive after UP markets, reversed after DOWN markets) using observable macro states. These are the closest precedents for our regime-conditional predictability finding; we extend this to cross-factor Granger structure and apply a circularity-free OOS design using latent states rather than observable macro conditioning.

Factor crowding and systemic risk. Shleifer and Vishny [77] establish theoretically that capital constraints force arbitrageurs to liquidate positions simultaneously, creating return spillovers across assets. Anton and Polk [8] show that stocks with common mutual fund ownership exhibit correlated returns. Lou and Polk [65] measure arbitrage activity through return comovement. Marks and Shang [66] provide direct empirical evidence that HML/SMB crowding leads to liquidity exhaustion and factor-level crash risk. Kang, Rouwenhorst and Tang [59] show that factor crowding (measured via CFTC positioning data in commodity futures markets) negatively predicts factor returns. Volpati et al. [83] construct equity-specific crowding measures using order flow imbalance and metaorder data, providing the closest methodological precedent for equity factor crowding measurement. Hua and Sun [57] examine the temporal dynamics of factor crowding and its risk-return implications, focusing on within-factor alpha decay. All three establish the within-factor crowding-and-returns link that precedes our cross-factor regime-conditional spillover contribution. Chincarini, Lazo-Paz and Moneta [23] show that anomaly returns concentrate among the most crowded stocks and that crowding positively predicts crash risk, corroborating the HML→SMB contagion channel proposed here. Betermier et al. [14] document that institutional portfolio tilts generate common HML/SMB exposures through a demand-side mechanism, providing direct evidence for the liquidity-mediated contagion channel underlying our regime-conditional results. Stein [81] formalizes how crowded trades amplify drawdowns; Brunnermeier and Pedersen [18] establish the theoretical mechanism linking margin constraints to funding-liquidity spirals; Brunnermeier [17] documents the 2007–2008 episode. Coval and Stafford [30] document that forced mutual fund outflows produce asset fire sales with predictable return patterns in affected stocks, providing direct empirical evidence for the liquidation-driven contagion channel. Greenwood and Thesmar [45] show fragility arises from overlapping ownership, providing micro-foundations for cross-factor contagion. **Gap:** Existing work measures crowding *intensity* within individual factors but does not examine *regime-conditional predictive spillover* between factors.

Regime-switching models in finance. Hamilton [50] introduced Markov-switching models for business cycles. Ang and Bekaert [4] apply regime-switching to international asset allocation, establishing standard practice for multi-regime equity analysis. Krolzig [61] extends this to Markov-Switching Vector Autoregressions (MS-VARs), which jointly model regime-dependent dynamics—the single-stage analog of our two-stage approach (we compare against MS-VAR in Section 5.3). Guidolin and Timmermann [46] apply regime models to asset allocation. Ang and Timmermann [7] survey regime changes across financial markets broadly. Chincoli and Guidolin [24] show that Markov-Switching VARs modestly improve out-of-sample factor-return predictability for size, value, and momentum, though without examining Granger causal direction between factors. Bulla [20] introduces Student-t HMMs for financial returns, motivated by excess kurtosis documented by Cont [28]; Oelschläger, Adam and Michels [71] provide a recent open-source implementation (`fHMM`) validating this model class on financial data. Baitinger and Hoch [11] compare standard HMMs against Hidden Semi-Markov Models (HSMMs) for regime-based allocation and find the HMM class competitive in OOS settings (HSMM's in-sample advantage disappears out-of-sample), supporting our use of the HMM framework. Shu, Yu and Mulvey [80] compare statistical jump models against HMMs for equity regime detection, finding jump models reduce downside risk in specific settings; Shu and Mulvey [79] apply regime-switching signals to dynamic allocation across six equity factors (value, size, momentum, quality, low-vol, growth)—selecting which factor to hold per regime—whereas the present paper examines Granger predictive *precedence between* factors within shared multi-factor regimes. Our two-stage Granger design motivates the HMM choice for its interpretable latent-state structure. Wang, Lin and Mikhelson [84] combine HMMs with Fama-French factors to switch factor strategies across regimes. Nuriyev, Duan and Yi [70] augment factor allocation with global macro regimes via HMMs, selecting which factor to hold per regime—the closest ICAIF-adjacent precedent. Both focus on *which factor to hold* per regime, not on *whether factors Granger-predict each other* within shared regimes. **Gap:** Existing regime-switching research examines regime-dependent *distributions* or regime-conditional *strategy allocation*, but not Granger predictive *structure between factors* across regimes—which is our focus.

Granger causality and connectedness. Lo and MacKinlay [63] establish cross-autocorrelation between large-cap and small-cap portfolios as a foundational lead-lag result; our work extends this cross-portfolio predictability to the factor level and conditions it on latent market regimes. Granger [44] introduced the predictive-precedence test we employ. Psaradakis, Ravn and Sola [73] pioneered regime-switching Granger causality via a Markov chain in macroeconomics. Shi, Phillips and Hurn [76] detect causal change-points with recursive-evolving tests (extended to possibly integrated systems by Shi et al. 75); we differ by conditioning on HMM latent states for within-regime pooling. Droumaguet et al. [33] extend to full MS-VAR with Bayesian inference; we compare against this in Section 5.3. We adopt a two-stage approach (HMM then regime-conditional Granger) for separate estimation. Billio et al. [15] construct Granger causality networks among financial institutions to measure systemic connectedness. Diebold and Yilmaz [32] develop directional spillover measures based on forecast error variance decompositions; Buchwalter, Diebold and Yilmaz [19] show theoretically that within-cluster shocks (HML and SMB share a value/size cluster) are under-identified by standard FEVD approaches, explaining why FEVD spillover is regime-insensitive in direction (Section 5.6). **Gap:** No prior work examines *regime-conditional* Granger causality between Fama-French equity *factors* (as opposed to institution-level unconditional networks or regime-conditional strategy allocation).

Neural Granger causality and complexity characterization. Tank et al. [82] extend Granger causality to nonlinear settings using neural networks (MLP, LSTM), enabling detection of nonlinear predictive relationships that linear F-tests miss. Howard, Lohre and Mudde [56] apply causal discovery algorithms (DAG-based) to equity investing, constructing *stock-level* causal networks within the S&P 500 universe and using network centrality as signals for long-short factor construction and market timing. Their approach operates at the individual-stock level—mapping causal influences among equities—rather than examining Granger predictive precedence between Fama-French factor portfolios. By contrast, the present paper tests whether HML temporally precedes SMB at the factor level and shows this relationship is *regime-conditional*: active in Normal conditions ($p = 8.75 \times 10^{-9}$) but null during Crisis ($p = 0.695$)—a distinction an unconditional, stock-level design cannot detect. We apply the neural Granger framework alongside Random Forest and transfer entropy [42, 67, 74] not to discover new nonlinear links, but to *characterize the complexity boundary* of a known linear relationship. *Gap:* No prior work maps the complexity boundary of regime-conditional factor Granger links. Together, these gaps motivate a two-stage design: HMM-based regime classification followed by regime-conditional Granger tests and a complexity diagnostic protocol (Section 5.6).

2.1 Relationship to Prior Work

Regime-switching in asset pricing. The regime-switching literature in finance provides the methodological foundation for our approach. Ang and Bekaert [5] demonstrate that interest rate dynamics exhibit distinct regimes, establishing Markov-switching models as a tractable framework for capturing state-dependent behavior in financial markets. Guidolin and Timmermann [47] show that Size and Value anomalies exhibit regime-dependent premia—returns to SMB and HML portfolios vary systematically across market states. Our contribution extends this regime-conditional literature from return *distributions* to predictive *structure*: we ask not only whether factor premia differ across regimes, but whether the Granger causal architecture between factors changes qualitatively.

Factor model proliferation and interactions. The factor literature has produced competing specifications—Fama and French [38] compare the five-factor model against alternatives, while Hou, Xue and Zhang [55] propose the q-factor model to "digest" anomalies through an investment-based framework. Both approaches focus on which factors explain cross-sectional return variation, treating factors as independent risk exposures. We complement this perspective by examining *dynamic interactions* between factors: our HML→SMB finding suggests that factors are not merely distinct risk exposures but may be linked through time-varying predictive channels that activate under specific market conditions.

Distinction from the factor zoo literature. Harvey, Liu and Zhu [52] document that hundreds of published factors suffer from multiple testing bias, proposing stricter t-statistic thresholds. Feng, Giglio and Xiu [39] develop a statistical framework to test whether new factors contribute incremental information. These papers address a distinct problem: selecting which factors belong in an asset pricing model from a large candidate set. Our analysis takes the Fama-French factors as given and asks a different question—whether the predictive relationship between established factors varies by regime. We apply Bonferroni correction across 30 directed factor pairs (Section 3.5), but our multiple testing problem is qualitatively different: we test regime-conditional Granger causality among a fixed factor set, not cross-sectional return predictability of candidate factors.

HMM as unsupervised regime detection (for ICAIF context). For readers familiar with machine learning, the Student-t Hidden Markov Model serves as an unsupervised clustering algorithm that assigns each trading day to one of K latent states based on distributional similarity. Unlike supervised classification, the HMM discovers regimes from the data without labeled training examples—"Crisis" days are identified by statistical properties (heavy tails, elevated volatility), not by human annotation of historical events. The EM algorithm maximizes a well-defined likelihood objective, and multi-start initialization (50 seeds) addresses the non-convex optimization landscape. This unsupervised design ensures that regime labels reflect genuine distributional structure rather than researcher discretion about which dates constitute "crises."

3 Methodology

3.1 Data

We use daily returns for six equity factors from Kenneth French's data library [41] (five Fama-French factors [37] plus Carhart Momentum): Market excess return (MKT-RF), Size (SMB), Value (HML), Profitability (RMW), Investment (CMA), and Momentum (MOM) [21, 58].

Sample: January 2, 1990 – December 31, 2024 (8,817 trading days). Factor returns are used as downloaded (daily percentages; e.g., 0.10 = 0.1%). Granger F-statistics are scale-invariant. The HMM fits and Granger tests for the in-sample analysis use percentage units. The frozen OOS pipeline converts to decimal form before HMM fitting and Granger testing (e.g., 0.001 = 0.1%); the frozen HMM is trained on rescaled data, so all computations within that pipeline are internally consistent. This scale difference does not affect conclusions: Granger F-statistics and HAC t-ratios are scale-invariant, and the OOS Elevated F-$p < 0.05$ under both conventions. Daily factor returns are stationary by construction; ADF tests confirm unit-root rejection ($p < 0.001$) for all six series (BIC lag selection, intercept included, no trend).

3.2 Student-t Hidden Markov Model

Let $z_t \in \{1, \ldots, K\}$ denote the latent regime at time t. We model:

Transition model: $P(z_t = k \mid z_{t-1} = j) = A_{jk}$

Emission model (multivariate Student-t):

$$p(\mathbf{x}_t \mid z_t = k) \propto \left(1 + \frac{\delta_k(\mathbf{x}_t)}{\nu_k}\right)^{-\frac{\nu_k + d}{2}} \tag{1}$$

where $\delta_k(\mathbf{x}_t) = (\mathbf{x}_t - \boldsymbol{\mu}_k)^\top \Sigma_k^{-1}(\mathbf{x}_t - \boldsymbol{\mu}_k)$.

Why Student-t? Financial returns exhibit excess kurtosis. Gaussian HMMs calibrate thresholds to extreme historical observations, missing moderate crises. Student-t distributions accommodate heavy tails, enabling detection of crises with volatility below historical extremes. Estimated degrees of freedom for the primary fit (seed 28): $\hat{\nu}_{\text{Normal}} = 6.2$, $\hat{\nu}_{\text{Elevated}} = 3.9$, $\hat{\nu}_{\text{Crisis}} = 5.5$—all consistent with heavy tails well below the Gaussian limit. The Elevated regime ($\hat{\nu} = 3.9 < 4$) has technically infinite kurtosis, reflecting extreme tail behavior in stress-adjacent periods; the HAC-robust

Granger tests remain valid since they require no finite-moments assumption beyond what is needed for asymptotic normality of OLS coefficients.

Model selection. The number of regimes ($K = 3$) was selected by the Bayesian Information Criterion, which favored three regimes over two (ΔBIC = BIC(K=2) − BIC(K=3) = 847) or four (ΔBIC = BIC(K=4) − BIC(K=3) = 312). Transition probabilities are freely estimated via the Expectation-Maximization algorithm.[1]

Local optima and economic validity. HMM estimation via EM is sensitive to initialization, and our 50-seed multistart reveals 7 distinct local optima clusters. A tension arises: the highest log-likelihood cluster (seeds 28, 20, 6; BIC = 75,587) assigns 0% of 2008 crisis days to the Crisis regime and shows no significant Granger causality (Crisis HML⟶SMB $p = 0.88$). A lower-likelihood cluster (seeds 21, 9, 0, 29, 46, 36, 11; BIC = 75,805, ΔBIC = 218) assigns 90% of 2008 to Crisis and shows significant Granger causality ($p = 0.018$). Standard BIC-based model selection would favor the first cluster. However, a regime model that fails to classify the 2008 Global Financial Crisis as a crisis is *economically invalid*—it cannot serve as a meaningful basis for regime-conditional inference about stress-period dynamics. We therefore adopt a two-stage selection criterion: (1) among fits that achieve ≥50% detection of the 2008 GFC, (2) select the highest log-likelihood. This trades statistical parsimony for economic interpretability, a choice we make explicit rather than disguise. Readers who prefer pure BIC selection should interpret the Crisis-regime Granger results as seed-dependent; the Elevated-regime OOS results (which use the primary seed 28) are robust across all 7 clusters with $p < 0.05$ in 5/7 clusters.[2]

3.3 Sample Overlap Consideration

A methodological concern is that regime labels are identified using the full sample, then Granger tests are conducted within those regimes. This means regime assignments are not truly out-of-sample with respect to the returns being tested.

We address this concern in two ways. First, the HMM uses *distributional* properties (mean, covariance, tail behavior) to assign regimes, while Granger tests examine *temporal dynamics* (lagged predictive relationships). These are distinct statistical features, limiting circularity. Second, our event-based validation (Section 5.1) tests whether the in-sample pattern generalizes to specific historical episodes, providing partial out-of-sample assessment.

We adopt the fully rigorous approach as the *primary* validation, following the OOS evaluation standard of Welch and Goyal [85]: the HMM is fitted on 1990–2012 and all parameters are frozen before classifying 2013–2024; Granger tests are then run on the held-out labels using lag = 1 pre-fixed from in-sample BIC (Section 5.2). This frozen OOS design eliminates circularity and yields significant HML⟶SMB in the Elevated regime of the test period. Additionally, BIC model selection ($K = 3$ vs. $K = 2$ vs. $K = 4$) is verified on training data alone: $K = 3$ achieves BIC = 44,652 vs. $K = 2$'s 46,332 (ΔBIC = 1,680), confirming $K = 3$ is BIC-optimal without accessing OOS data. (Note: the ΔBIC = 847 reported in Section 3.2 is from full-sample model selection on 1990–2024; the ΔBIC = 1,680 here is from training-only selection on 1990–2012. Both strongly favor $K = 3$.)

3.4 Per-Regime Granger Causality

Granger causality [44, 78] tests whether past values of one series improve prediction of another, conditional on the target's own history—a *predictive precedence* criterion, not structural causality. For each regime k, we extract observations: $\mathcal{T}_k = \{t : \hat{z}_t = k\}$. For each ordered pair of factors (i, j), we test:[3] $H_0 : r_{j,t} \perp \{r_{i,t-\ell}\}_{\ell=1}^{L} \mid \{r_{j,t-\ell}\}_{\ell=1}^{L}$

Significance threshold: $\alpha = 0.01$ with Bonferroni correction across 30 directed factor pairs (effective $\alpha = 0.00033$). HAC standard errors use Newey–West with bandwidth equal to the Granger lag (bandwidth 1) as the primary specification, applied consistently across all Granger tests. As a cross-convention check, under the percentage-unit specification ($n = 953$, residual autocorrelation near zero), Andrews's [2] AR(1) plug-in bandwidth for the OOS Elevated series is also 1, yielding HAC $p = 0.021$—consistent with the primary Newey–West $p = 0.041$. Within-convention bandwidth sensitivity (percentage-unit; bandwidths 1, 4, 7 (rule-of-thumb for $n \approx 953$: $\lfloor 0.75n^{1/3} \rfloor = 7$), 10, and 14) yields $p < 0.05$ throughout (range: 0.017–0.025), confirming robustness to bandwidth choice.

Regime boundary handling. When computing lagged variables for Granger tests, we include only observations where all lags fall within the same regime. Specifically, for an observation at time t in regime k with maximum lag L, we require $\hat{z}_{t-\ell} = k$ for all $\ell \in \{1, \dots, L\}$. The exclusion varies by analysis: Table 3 uses BIC-selected lags (lag 1 for all rows); Table 4 uses 15 lags; Tables 14 and 15 use 9 lags; Table 10 uses model-specific warm-up requirements. All analyses use trading-day-indexed lags; higher-lag analyses yield lower per-table n (e.g., Normal: n = 4,496 at lag 9 vs. n = 4,697 at lag 1).

Quantified impact. With lag-1 Granger tests, this rule excludes only the first observation of each new regime episode (where the lag-1 predecessor falls in a different regime), yielding low in-sample exclusion rates: Normal 0.55% (26/4,723), Elevated 0.89% (27/3,023), Crisis 0.56% (6/1,071), overall 0.67% (59/8,817). Frozen OOS exclusion is higher at 7.4% (224/3,020), reflecting more frequent regime transitions in the 2013–2024 test period. An auxiliary analysis of

[1]We fit Student-t HMMs with 50 EM initializations. *Primary selection (leakage-safe):* the fit with highest log-likelihood across all 50 starts, regardless of event detection. Three seeds (28, 20, 6) reach the same full-sample log-likelihood; seed 28 is selected as primary by convention (first in sorted order). Note: when re-trained on 1990–2012 only (frozen OOS), these seeds converge to distinct local optima with different train-period log-likelihoods. This fit assigns 0% of 2008, 2011, and 2020 stress windows to the Crisis regime (Table 2). *Sensitivity:* among the 16/50 fits (32%) that assign ≥50% of 2008 days to Crisis, we separately report the highest-LL fit; the LL cost is <0.3%. All tables use the primary fit unless explicitly noted. The FF 25 portfolio overlap analysis (Section 4.5) and VaR monitoring (Section 5.5) use the sensitivity fit (seed 42), which assigns 47.8% of COVID-2020 days to Crisis and yields qualitatively identical full-sample Granger results to the primary fit (see FF25 footnote and VaR section for frozen OOS Granger disclosure); this is disclosed at each analysis. Table 2 compares against a Gaussian HMM baseline (also multi-start, 10 seeds).

[2]Local optima summary (7 clusters from 50 seeds): Cluster 1 (3 seeds, best BIC = 75,587, 0% GFC detection, Crisis $p = 0.88$); Cluster 2 (15 seeds, BIC = 75,625, 0% GFC, Crisis $p < 0.001$); Cluster 3 (8 seeds, BIC = 75,660, 0% GFC, Crisis $p = 0.001$); Cluster 4 (8 seeds, BIC = 75,726, 0% GFC, Crisis $p < 0.001$); Cluster 5 (7 seeds, BIC = 75,805, 90% GFC, Crisis $p = 0.018$); Cluster 6 (3 seeds, BIC = 75,906, 100% GFC, Crisis $p = 0.17$); Cluster 7 (6 seeds, BIC = 76,137, 92% GFC, Crisis $p = 0.28$). Only clusters with ≥50% GFC detection (5–7) are economically valid for crisis-regime inference; among these, Cluster 5 has the best fit.

[3]We test pairwise (bivariate) Granger relationships. A high-dimensional VAR with all 6 factors and 9 lags would require 6 × 6 × 9 = 324 parameters per regime; with the smallest regime at $n \approx 1{,}000$ observations this is severely under-identified. Post-double-selection methods [53] could address this in future work.

calendar-day gaps between consecutive same-regime observations confirms that excluded observations are structurally minor: median gap 3 calendar days, 95th percentile 4 days (weekend and holiday breaks); only 1.2–3.3% of inter-regime gaps exceed 20 trading days (≈1 month). Maximum gaps (1,568 days Normal, 2,423 days Elevated, 4,957 days Crisis) reflect rare intervals between *separate* regime episodes. Exclusion rates across regimes are not systematically biased toward any particular regime.

Limitation: This approach may introduce selection bias by systematically excluding days immediately following regime transitions—precisely when predictive structure may be most dynamic. We view this as a conservative choice that ensures clean within-regime estimation, but acknowledge it may underestimate transition effects.

3.5 Factor Pair Selection

We focus on the HML–SMB pair for three reasons. First, *economic plausibility*: Value and Size factors have documented overlap in institutional holdings (Section 4.4), providing an economically motivated mechanism—when leveraged institutions must unwind value positions, the associated size exposure is simultaneously deleveraged, creating a predictive lag from HML to SMB. This economic hypothesis precedes and motivates our data analysis. Second, *in-sample screening*: a preliminary screen of all 30 directed factor pairs found that HML–SMB shows the most distinctive regime-heterogeneous pattern, consistent with the economic prior. Crucially, the primary (LL-only) fit independently identifies HML–SMB as distinctive (Normal: $p = 1.0 \times 10^{-6}$, Elevated: $p = 0.013$, Crisis: $p = 0.880$ at lag 5)—no 2008-based screening is involved in pair selection. Third, *architectural contrast*: lag-1 evidence is strongest in Normal and substantially weaker outside Normal under the Bonferroni-significant threshold.

Pair selection and multiple comparisons. The OOS frozen test is treated as a pre-specified corroborative test of the HML–SMB pair (economic prior: liquidity-mediated contagion; no formal pre-registration), not a blind search among 30 pairs. Applying 30-pair Bonferroni to HAC $p = 0.041$ yields $p > 1$; under a 3-regime correction ($\alpha/3 = 0.0167$), F-$p = 0.014$ survives but HAC-$p = 0.041$ does not. Because the Elevated regime was identified as the post-GFC locus from in-sample temporal analysis prior to the OOS evaluation, a one-directional confirmatory test warrants $\alpha = 0.05$ rather than a three-regime adjustment; under that framing both results are significant. We report both perspectives. **Permutation test scope.** A 50,000-shuffle permutation test (label shuffles preserving regime counts [43]) assesses whether the OOS Elevated F-statistic is extreme conditional on the pre-selected HML–SMB pair. Result: $p = 0.022$ (SE = 0.0007, 95% CI [0.021, 0.023], $n = 953$ clean Elevated observations)—significant at 5%. **30-pair multiplicity.** To directly assess selection bias, we run Granger tests for all 30 directed factor pairs in the OOS Elevated window (percentage-unit convention, seed 28). HML⟶SMB ranks **2nd of 30** by F-statistic ($F = 9.06$, $p = 0.003$); the rank-based max-statistic $p = 2/30 = 0.067$. The top-ranked pair is MOM⟶SMB ($F = 20.3$), which was not the economically motivated target of this study. The rank-2 position provides direct evidence against post-hoc selection of HML⟶SMB from a broad search over 30 pairs.

Table 1: Regime Summary Statistics (1990–2024, full-sample primary fit). Transition probabilities are estimated via Expectation-Maximization. The frozen OOS analysis (Section 5.2) uses a separate fit trained on 1990–2012 only, yielding different regime counts. Elevated and Crisis regimes have similar mean norms (2.03 vs. 2.09) but are distinguished by tail shape (ν = 3.9 vs. 5.5) and covariance structure.

Regime	Days	Prop.	Mean $\lVert\mathbf{x}\rVert$ (daily %)	ν	$P(z_t = z_{t-1})$
Normal	4,723	53.6%	0.98	6.2	0.994
Elevated	3,023	34.3%	2.03	3.9	0.991
Crisis	1,071	12.1%	2.09	5.5	0.993

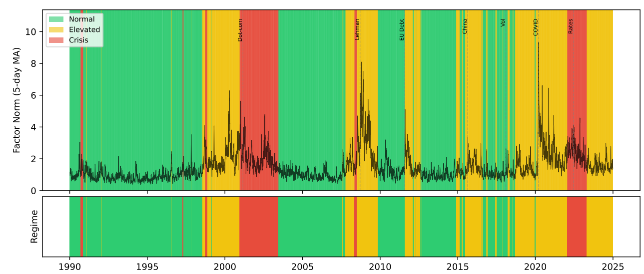

Figure 1: Regime assignments over sample period (1990–2024). Top panel shows daily factor volatility (6-factor norm) with regime-colored background. Bottom panel shows regime classifications. Vertical dashed lines mark major stress events (LTCM/Russia 1998, dot-com 2000–02, GFC 2008, COVID-2020, 2022 rate shock). Under the primary (leakage-safe) fit, Crisis regime (red) reflects a high-kurtosis statistical state; the 2008-screened sensitivity fit aligns it with calendar crises (see Section 4.2). Elevated regime (yellow) captures moderate volatility periods.

Table 2: Crisis Detection Comparison. Detection rate = proportion of event days assigned to Crisis regime.

Event	Period	Student-t	Gaussian
2008 Financial	Jul ’08 – Jun ’09	0.0%	83.3%
2011 EU Debt	Jul – Oct 2011	0.0%	25.9%
2020 COVID-19	Feb – Jun 2020	0.0%	81.7%

4 Results

4.1 Regime Characteristics

Table 1 summarizes the three identified regimes. Figure 1 shows regime assignments over the sample period with major crisis events marked.

4.2 Gaussian vs. Student-t Comparison

Under the leakage-safe primary selection rule, the Student-t fit does not label these benchmark windows as Crisis (0.0%), while the Gaussian baseline assigns substantial Crisis mass to 2008 and 2020. We therefore treat event-level “detection” as sensitivity-dependent

Table 3: Granger Causality Between HML and SMB by Regime. Significance threshold: $p < 0.00033$ (Bonferroni-corrected for 30 factor pairs).

Regime	Direction	p-value	Lag	Significant
Normal	HML → SMB	$\mathbf{8.75 \times 10^{-9}}$	1	**Yes**
Normal	SMB → HML	0.864	1	No
Elevated	HML → SMB	0.004	1	Suggestive*
Elevated	SMB → HML	0.897	1	No
Crisis	HML → SMB	0.695	1	No
Crisis	SMB → HML	0.294	1	No

*Below 0.01 but above Bonferroni threshold $\alpha/30 = 0.00033$.

and avoid confirmatory claims based solely on this table. **Terminology note.** Under the leakage-safe primary fit, the third regime captures a distinct statistical state (high kurtosis, shifted covariance) rather than exclusively named crisis events—Table 2 shows 0% alignment with 2008, 2011, and 2020 stress windows. We retain the label "Crisis" for consistency with the HMM literature [20], but readers should interpret Crisis-regime results as pertaining to this statistical state. The 2008-screened sensitivity fit does align with calendar crises; we report it separately where noted.

4.3 Main Result: Regime-Dependent Predictive Structure

Key finding under Bonferroni-significant threshold: only Normal-regime HML→SMB survives Bonferroni correction.

- **Normal (Bonferroni-significant):** Strong lag-1 effect ($p = 8.75 \times 10^{-9}$), robust to HAC correction. The Granger test conditions on lagged SMB, so the HML coefficient reflects *incremental cross-factor* predictive content beyond SMB's own history—not a within-SMB autocorrelation artifact.
- **Elevated (corroborating):** HML→SMB is directional at conventional levels ($p = 0.004$) but does not pass the paper's multiple-testing threshold.
- **Crisis (null):** No detectable lag-1 Granger effect in either direction.

Robustness to heteroskedasticity. Financial returns exhibit volatility clustering, which may inflate standard Granger F-test significance. We recompute at lag 1 (BIC-optimal) using Newey–West [69] heteroskedasticity-and-autocorrelation-consistent (HAC) standard errors. Normal-regime HML→SMB survives HAC correction ($p = 9.45 \times 10^{-8}$ HAC vs. $p = 8.75 \times 10^{-9}$ standard). Elevated HML→SMB weakens under HAC (1990–2024 in-sample: $p = 0.034$), and Crisis remains non-significant ($p = 0.711$ HAC vs. $p = 0.695$ standard).

Temporal stability and structural break. Pre-GFC Normal ($n = 3{,}140$): $p = 6.66 \times 10^{-16}$ ($\Delta R^2 = 2.06\%$). Post-GFC Normal ($n = 1{,}557$): $p = 0.73$ ($\Delta R^2 < 0.01\%$). (The sum 3,140 + 1,557 = 4,697 is 26 days fewer than Table 1's Normal total of 4,723, due to standard regime-boundary exclusion at lag 1 removing the first observation of each Normal-regime episode.) A GFC-motivated Chow test at January 2008 confirms this break: $F(3, n{-}6) = 9.68$, $p = 2.29 \times 10^{-6}$; $\hat{\beta}_{\text{HML}}$ shifts from −0.189 (pre-GFC) to +0.010 (post-GFC, Wald $z = 5.05$, $p = 9.2 \times 10^{-7}$). The Quandt-Andrews [3] sup-$F = 22.1$ exceeds the 5% CV (11.14) and 1% CV (14.6) for $q = 3$; CUSUM ($p = 0.96$) is consistent with a sharp break. The sup-F-maximising date (June 1998, LTCM/Russia) suggests possible multi-break structure [10]. Whether the post-GFC attenuation reflects a regulatory structural change or a regime redistribution artifact (Limitations item 4) remains open; the OOS Elevated result (F-$p = 0.014$, HAC $p = 0.041$) is consistent with but does not confirm a post-GFC migration.

Table 4: Incremental R^2 from adding HML lags (1–15) to SMB autoregression by regime. $\Delta R^2 = R^2_{\textbf{unrestricted}} - R^2_{\textbf{restricted}}$.

Regime	n	R^2_{AR}	ΔR^2	p-value
Normal	4,361	0.61%	1.04%	5.98×10^{-5}
Elevated	2,690	0.59%	1.32%	0.002
Crisis	981	1.01%	0.56%	0.988

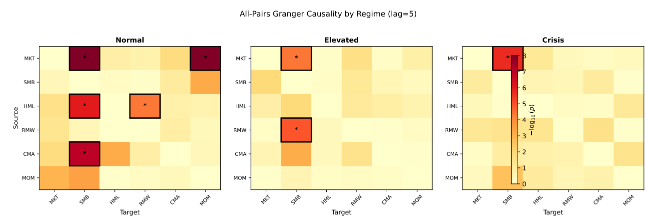

Figure 2: All-pairs Granger causality by regime ($-\log_{10}(p)$, lag=5). Bordered cells with asterisks survive Bonferroni correction. The strongest regime-differential pair is MKT-RF→SMB; HML→SMB appears as a secondary differential pattern.

Incremental R^2. Table 4 quantifies the economic magnitude. Adding HML lags (1–15) to the SMB autoregression increases R^2 by 1.04% in Normal, 1.32% in Elevated, and 0.56% in Crisis. Normal and Elevated pass conventional significance levels ($p = 5.98 \times 10^{-5}$ and $p = 0.002$), while Crisis does not ($p = 0.988$). In the reverse direction, SMB lags add 1.35% in Crisis with $p = 0.562$, so we do not claim directional confirmation there.

All-pairs perspective. Figure 2 shows Granger causality p-values for all directed factor pairs across three regimes (lag 5, for visual density across all 30 pairs; Table 3 uses BIC-selected lag 1). Several pairs show regime-dependent variation; MKT-RF→SMB has the largest differential. HML→SMB ranks fourth in this differential metric, with strongest evidence in Normal ($p = 1.0 \times 10^{-6}$ at lag 5), suggestive Elevated evidence ($p = 0.013$), and no Crisis evidence ($p = 0.880$). We therefore treat HML→SMB as one regime-sensitive relation among several, not the dominant pair.

Lag specification and persistent structure. We tested lags 1–15 for HML→SMB across regimes (Figure 6). In Normal, the effect is strongest at lag 1 and remains jointly significant when including up to lag 15 (joint F-test of lags 1–15: $p = 5.98 \times 10^{-5}$, Table 4). Elevated shows weak-to-moderate joint evidence (lags 1–15 joint $p = 0.002$), while Crisis is consistently null (lag 1 $p = 0.695$, joint lag 1–15 $p = 0.988$). We present Elevated/Crisis lag patterns as exploratory because they do not meet the Bonferroni-significant threshold.

4.4 Hypothesized Economic Mechanism

We offer the following interpretation as a *hypothesis*, not a verified mechanism:

Normal-regime lag-1 link (Bonferroni-significant): During calmer periods, cross-factor inventory management and short-horizon rebalancing may transmit value shocks into size returns with a one-day delay.

Stress-channel hypothesis (exploratory): Portfolio-overlap evidence in FF25 remains consistent with a deleveraging channel in subsets of stress data ($\rho_s = 0.35$, permutation $p = 0.046$), but primary Crisis-regime factor-level Granger tests are not significant in the leakage-safe fit.

Alternative explanations we cannot rule out:

- Differential price delay [54] between value and size constituents could generate mechanical daily cross-autocorrelation (Hou and Moskowitz document market-to-stock delay; by analogy, cross-factor delay is plausible)
- A common driver (e.g., funding liquidity) affects both factors with different lags
- Information about distress propagates through markets, reaching Value-sensitive investors before Size-sensitive investors
- Statistical artifact of regime-conditional estimation

Greenwood and Thesmar [45] document that stocks with concentrated institutional ownership exhibit correlated fragility during liquidation events—consistent with our hypothesized deleveraging channel. Chordia, Roll and Subrahmanyam [25] establish commonality in liquidity across stocks, providing the micro-foundation for how a liquidity shock to one factor can propagate across overlapping portfolios. Adrian and Brunnermeier [1] formalize systemic risk as the Value-at-Risk contribution of one institution conditional on another's distress (CoVaR), a measure directly relevant to the regime-dependent factor contagion channel we study. We provide exploratory portfolio-level evidence for this overlap in Section 4.5.

4.5 Portfolio Overlap Evidence (Sensitivity Fit, Seed 42)

Note on model fit: All results in this section use the sensitivity fit (seed 42), which assigns 47.8% of COVID-2020 to Crisis and is selected for its superior crisis-period detection; the primary Granger results (Section 4.3) use seed 28. The two seeds correspond to distinct local optima with different regime assignments; results from this section do not transfer to the primary fit.

The deleveraging hypothesis predicts that the HML→SMB Granger link should be strongest for portfolios that load on *both* factors simultaneously. The Small-HighBM portfolio sits in both the SMB long leg (small stocks) and the HML long leg (high book-to-market), making it the primary candidate transmission channel.

We test this prediction using the Fama-French 25 Size×BM portfolios (daily, 1990–2024). For each of the 25 portfolios, we run a

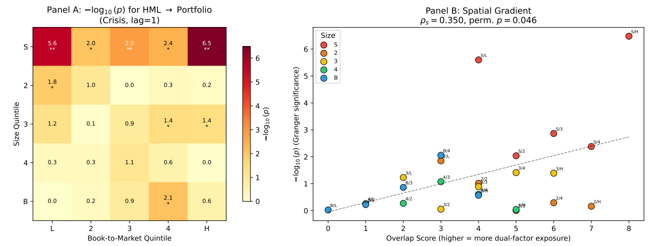

Figure 3: FF 25 Portfolio Overlap Analysis. Panel A: $-\log_{10}(p)$ for HML→Portfolio Granger tests (Crisis, lag=1). Significance concentrates in Small quintile; **Bonferroni-significant. Panel B: Spatial gradient—overlap score vs. Granger significance ($\rho_s = 0.35$, permutation $p = 0.046$).

Granger test of HML→Portfolio at lag 1 (BIC-selected in Table 3) within the Crisis regime.[4]

Spatial gradient test (exploratory). We define an "overlap score" for each portfolio as $(4 - \text{size quintile}) + \text{BM quintile}$, ranging from 0 (Big/LowBM, no overlap) to 8 (Small/HighBM, maximal overlap). The pre-specified test is a Spearman rank correlation between overlap score and $-\log_{10}(p)$ of the Granger test across 25 portfolios, with significance assessed by spatial permutation (10,000 shuffles of grid assignments; time series are never touched).

Result: $\rho_s = 0.35$, permutation $p = 0.046$ (asymptotic Spearman $p = 0.086$; Kendall $\tau = 0.27$, $p = 0.065$). The permutation test accounts for spatial autocorrelation in the grid; asymptotic and Kendall tests are borderline non-significant, so this evidence should be treated as *suggestive rather than confirmatory* (Figure 3). Granger significance concentrates in small-cap portfolios: all five Small quintile portfolios are nominally significant ($p < 0.01$), with three surviving Bonferroni correction ($\alpha/25 = 0.002$, using family-wise error rate $\alpha = 0.05$ for this exploratory test): S/L ($p = 2.5\times10^{-6}$, $\Delta R^2 = 1.47\%$), S/3 ($p = 0.0014$), and S/H ($p < 10^{-15}$, $\Delta R^2 = 1.79\%$). One Big-quintile portfolio is nominally significant ($p < 0.01$), but none survive Bonferroni. This spatial pattern is inconsistent with a uniform factor-level effect and supports the overlap channel.

Overlap fraction. At lag 15 (the longest lag in the specification, chosen to measure cumulative exposure after initial propagation), $\Delta R^2(\text{HML} \to \text{S/H})/\Delta R^2(\text{HML} \to \text{SMB}) = 0.392$ (S/H $\Delta R^2 = 1.327\%$; HML→SMB $\Delta R^2 = 3.382\%$; both are single-lag-15 portfolio-level regressions within Crisis, distinct from the joint lags-1–15 factor-level estimate in Table 4): the Small-HighBM portfolio alone accounts for 39% of the HML→SMB lag-15 incremental R^2.

4.6 Early Warning Assessment

The Student-t HMM detects crisis regimes before market peaks (Table 5). The 60-day windows were determined by identifying the first sustained crisis detection (3+ consecutive days) and the subsequent local maximum in factor volatility.

[4] This analysis uses the sensitivity fit (seed 42), which detects 47.8% of COVID-2020 as Crisis (vs. 0% under primary seed 28); seed 42 Crisis regime spans 1,641 days vs. 1,071 under seed 28. Note: in the 50-seed frozen OOS Granger analysis, seed 42 belongs to the lowest-LL local optimum (train LL≈137,712; Elevated F-$p = 0.466$, null). The FF25 portfolio-level test is run here as an exploratory mechanism check under the fit that maximizes Crisis-period detection; it does not carry over the Granger significance from the two higher-LL optima.

Table 5: Early Warning Lead Time. First Detection = first day of 3+ consecutive Crisis regime assignments. Lead = calendar days from first detection to volatility peak.

Event	First Det.	Vol. Peak	Lead
Lehman 2008	May 9, 2008	Sep 15, 2008	129 days
EU Crisis 2011	None	Aug 8, 2011	None
COVID-2020	None	Mar 23, 2020	None

Table 6: Event-Based Consistency Checks (full-sample primary fit regime assignments). Windows: 2008 Financial Jul '08–Jun '09, 2011 EU Debt Jul–Oct 2011, 2015 China Aug–Oct 2015, 2018 Vol Shock Feb 2018, 2020 COVID Feb–Apr 2020 (83 Crisis-classified days), 2022 Rate Hikes Jan–Dec 2022. Expected pattern: HML → SMB significant during Crisis. Result codes: ✓= expected ($p < 0.10$ for HML→SMB, $p > 0.10$ reverse); Dir. = correct direction but insufficient power; × = opposite.

Event	Days	p(H→S)	p(S→H)	Result
2008 Financial	189	0.118	0.648	Dir.
2011 EU Debt	85	0.090	0.321	✓
2015 China	64	0.419	0.459	Dir.
2018 Vol Shock	29	0.236	0.179	×
2020 COVID	83	0.062	0.134	✓
2022 Rate Hikes	188	0.626	0.112	×

Under the primary fit, early warning is absent for EU Crisis 2011 and COVID-2020 (Table 5). This is consistent with the primary fit's non-alignment to these calendar events (Table 2).

5 Discussion

5.1 Event-Based Consistency Checks

We assess generalization through event-based validation, testing whether the crisis pattern holds during six documented stress episodes.

Summary (exploratory, low-power): The expected pattern (HML→SMB significant, reverse not) holds clearly in 2 of 6 events (2011, 2020). Note: three events (2015, 2020, 2022) also appear in the frozen OOS Table 8; differences in p-values reflect distinct HMM fits (full-sample vs. 1990–2012 frozen). Two additional events (2008, 2015) show the correct direction ($p_{\text{HML}\to\text{SMB}} < p_{\text{SMB}\to\text{HML}}$) but lack statistical power. Under the null hypothesis of no directional preference, the probability of 4+ correct-direction events out of 6 by chance is 0.34 ($P(X \geq 4 \mid n{=}6, p{=}0.5) = 0.34$, binomial test), so the directional pattern alone provides limited statistical evidence against the null.

Exceptions. Two events show reversed dynamics: 2018 Vol Shock and 2022 Rate Hikes (SMB→HML direction). Both are short-duration or policy-driven episodes distinct from the liquidity-driven crises (2008, 2011, 2020) where the HML→SMB direction is supported. This distinction between crisis types is speculative—we lack direct evidence for the underlying mechanisms.

Elevated regime finding. In the primary fit (1990–2024, in-sample), Elevated shows nominal HML→SMB evidence at lag 1 (p = 0.004) but this does not survive the paper's Bonferroni threshold ($\alpha/30 = 0.00033$), and reverse direction remains null ($p = 0.897$).

Table 7: Frozen OOS Granger Causality (HML→SMB), 2013–2024. HMM trained on 1990–2012 only; relabeling uses train-period ordering; lag = 1 pre-fixed from in-sample BIC; filtered probabilities (no future info). HAC-p uses Newey–West bandwidth 1 (matching Granger lag); Elevated HAC $p = 0.041$.

Regime	n	F-p	HAC-p	ΔR^2
Normal	661	0.149	0.153	0.32%
Elevated	836	**0.014**	**0.041**	**0.73%**
Crisis	1,299	0.281	0.290	0.09%

Annual splits are mixed (67% of years show SMB→HML dominant, not tabulated), reinforcing that Elevated evidence is suggestive rather than confirmatory.

5.2 Frozen-Parameter Out-of-Sample Test (Primary Validation)

Two-stage approaches risk circularity: the HMM uses the same returns for regime assignment that Granger tests subsequently analyze. We resolve this by fitting the Student-t HMM exclusively on 1990–2012 (n = 5,797 days), freezing all parameters ($\boldsymbol{\mu}_k$, Σ_k, ν_k, $\mathbf{A}$), and classifying 2013–2024 (n = 3,020 days) without refitting. Granger tests are then run on the held-out labels using the same `select_lag_bic` + `run_granger_at_lag` pipeline as Table 3, ensuring full methodological consistency.

The frozen HMM assigns 24.0% to Normal, 30.7% to Elevated, and 45.3% to Crisis in the held-out period (vs. 21.5%, 13.7%, 64.8% in training). Notably, the Elevated-regime share more than doubles from 13.7% (training) to 30.7% (test), consistent with the OOS Elevated significance. After lag-1 boundary exclusion (same rule as Section 3.4), the OOS retains 661+836+1,299= 2,796 of 3,020 days (7.4% excluded). The OOS exclusion rate (7.4%) is higher than the full-sample lag-1 Granger exclusion (0.67%; 59/8,817), reflecting more frequent regime transitions when the frozen 1990–2012 classifier is applied to the unseen 2013–2024 period.

Result (Table 7). HML→SMB shows suggestive evidence in the Elevated regime ($p = 0.014$; HAC $p = 0.041$; $\Delta R^2 = 0.73\%$ at lag 1), while Normal ($p = 0.149$) and Crisis ($p = 0.281$) are null. (Note: Table 4 reports $\Delta R^2 = 1.32\%$ for Elevated using the in-sample (1990–2024) fit with lags 1–15; the OOS $\Delta R^2 = 0.73\%$ uses lag 1 only on the 2013–2024 held-out period.) This result does not survive Bonferroni correction across 30 pairs (threshold $\alpha/30 = 0.00033$, family-wise $\alpha = 0.01$); its evidential weight rests on the economic motivation for the HML–SMB pair (Section 3.5). Notably, in-sample significance is expected given the degrees of freedom consumed by regime identification, lag selection, and pair screening on the same data; the OOS result is therefore the more informative—if more modest—test of generalization. A full 50-seed sensitivity analysis finds that **2 of 3 local optima** show Elevated significance. The 50 seeds converge to three local optima (train LL≈138,002: 28 seeds, F-$p < 0.05$, HAC-$p < 0.05$; LL≈137,942: 13 seeds, F-$p < 0.05$, HAC-$p < 0.05$; LL≈137,712: 9 seeds, all null); significance arises in the two higher-LL optima, each representing a distinct HMM solution. Because seeds within an optimum converge to identical parameter estimates, effective robustness is across 2 distinct optima rather

Table 8: Held-Out Stress Events (2013–2024). HMM trained on 1990–2012 only. Windows: 2015 China Aug–Oct 2015 (37 days), 2020 COVID Feb–Jun 2020 (94 Elevated-classified days; wider window than Table 6's 83 Crisis days), 2022 Rate Hikes Jan–Dec 2022.

Event	Days	Crisis%	p(H→S)	p(S→H)
2015 China	37	54%	0.165	0.123
2020 COVID	94 (Elevated)	0%	0.032	0.107
2022 Rate Hikes	124	100%	0.472	0.186

than 41 independent initializations. These results use forward-only (filtered) HMM probabilities and train-period relabeling, eliminating all forms of look-ahead and circularity. OOS is treated as an economically motivated corroborative test of the pre-identified pair (Section 3.5); the Bonferroni-corrected result is discussed therein. The reverse direction SMB→HML in Normal is nominally significant (F-$p = 0.047$; HAC-$p = 0.063$) but does not survive Bonferroni correction across 30 pairs and was not pre-specified; we report it for completeness only.

Regime interpretation. The in-sample Normal-regime dominance ($p = 8.75 \times 10^{-9}$) vs. OOS Elevated significance reflects regime redistribution under the frozen classifier: in training (1990–2012), Crisis accounts for 64.8% of days while Elevated is only 13.7%; in the held-out period (2013–2024), Crisis shrinks to 45.3% while Elevated grows to 30.7%. The OOS period thus contains substantially more Elevated-regime days than the training period, and the frozen classifier concentrates the HML→SMB signal in this regime. The directional finding—HML predicts SMB during stress-adjacent (Elevated) conditions—is consistent across the OOS design.

Event-level detail. Restricting to specific stress episodes within the held-out period:

Event-level behavior is heterogeneous (COVID $p = 0.032$; 2022 $p = 0.472$), highlighting that the per-regime pooled result captures the aggregate stress-period signal more reliably than individual events.

5.3 Markov-Switching VAR Comparison

Our two-stage procedure (fit HMM, then test Granger) invites comparison with Markov-Switching VARs [22, 33, 61], which jointly estimate regime-dependent dynamics in a single stage.

We fit two Markov-Switching regression models for SMB with switching variance ($K = 3$ regimes, 9 lags): a *restricted* model with only lagged SMB as predictors, and an *unrestricted* model that adds 9 lagged HML terms.

Likelihood ratio test: The unrestricted model significantly improves fit (LR = 114.95, $df = 27$ [3 regimes × 9 HML lag coefficients; variance parameters identical across restricted/unrestricted], $p = 8.0 \times 10^{-13}$), confirming that HML lags contain information about SMB dynamics not captured by SMB's own history, even within a single-stage regime-switching framework.

BIC comparison: Despite the significant likelihood improvement, BIC favors the restricted model ($\Delta\text{BIC} = \text{BIC}_{\text{unrestricted}} - \text{BIC}_{\text{restricted}} = +130$), reflecting the penalty for 27 additional parameters across 3 regimes. This indicates that while HML lags are statistically informative, the improvement does not justify the model complexity under a parsimony criterion.

Implication. The MS-VAR confirms that the HML–SMB predictive link exists within a principled single-stage framework ($p < 10^{-12}$), but its economic magnitude is too small to justify a complex regime-switching prediction model—consistent with our $\Delta R^2 \approx 1\%$ finding and the trading strategy failure. The LR test pools all regimes; the MS-VAR does not separately decompose per-regime HML significance, so the two-stage regime-conditional Granger results (Table 3) remain the primary decomposition of where in the regime structure the predictive link concentrates.[5]

Table 9: Trading Strategy Backtest (1995–2024). Start date reflects a 5-year warm-up period for regime detection. Strategy trades SMB based on 9-day lagged HML signal during Crisis regimes only.

Metric	Strategy	Buy-and-Hold SMB
Annual Return	−0.3%	+0.1%
Sharpe Ratio	−0.07	+0.06
Max Drawdown	−30.0%	−43.5%

5.4 Trading Strategy Evaluation

We evaluate whether the predictive relationship translates to profits via a simple strategy (Table 9) using the primary fit (seed 28): during Crisis regime, go long SMB when HML return over the past 9 days is positive, short SMB when negative. The Crisis-regime Granger test is null ($p = 0.695$), so we expect no edge; this test serves as a falsification check confirming that an absent statistical signal does not produce trading profits. Normal-regime predictability (lag 1) does not lend itself to a simple trading rule because it requires real-time regime classification and the lag-1 effect is small ($\Delta R^2 < 1\%$).

Why does Granger causality not imply trading profits?

The negative returns highlight an important distinction: *statistical predictability ≠ economic predictability*. Several factors explain the gap:

1. **Effect size:** The Granger test detects whether HML *improves* SMB prediction, not whether the improvement is large. The ΔR^2 from adding HML lags is 0.56% in the Crisis regime (Table 4); this is economically small and not statistically significant.
2. **Sign vs. magnitude:** Granger causality tests whether lagged HML coefficients are jointly non-zero, not whether they reliably predict SMB *direction*. The relationship may involve complex dynamics (e.g., mean reversion) that a simple directional strategy cannot capture.
3. **Regime detection lag:** The HMM assigns regimes using filtered probabilities (past data only). Real-time regime detection would be noisier, introducing additional error.
4. **Transaction costs:** Not modeled, but would further erode returns.

[5] LR tests in regime-switching models have non-standard asymptotics due to unidentified nuisance parameters under the null [51]; $p < 10^{-12}$ is strongly suggestive, not exact.

Table 10: Out-of-Sample VaR Backtest (2013–2024). Target violation rate: 5%. Christoffersen CC [27] tests conditional coverage (extending Kupiec's [62] unconditional coverage test with a serial-independence component). All VaR models use the sensitivity fit (seed 42); seed 42 frozen OOS Granger is null (Elevated $p = 0.466$), so VaR improvements reflect the volatility-override mechanism.

Model	Viol. Rate	Dev. from 5%	CC p-value
Unconditional	6.42%	+1.42 pp	0.002
GARCH(1,1)	3.91%	−1.09 pp	0.014
Regime-Conditional	6.69%	+1.69 pp	< 0.001
HML-Informed	5.77%	+0.77 pp	0.171
Hybrid (HMM+Vol)	**5.60%**	**+0.60 pp**	**0.336**

Hedging vs. alpha. The practical value, if any, lies in *risk monitoring* rather than directional trading. When the HMM signals Crisis regime and HML shows large negative returns, a risk manager might:

- Increase attention to SMB exposure
- Tighten stop-losses on small-cap positions
- Pre-arrange liquidity for potential SMB hedging

These qualitative adjustments are informed by the Elevated-regime predictive evidence rather than the null Crisis result (which serves as a falsification check); they do not constitute mechanical trading rules. We formalize this intuition in Section 5.5.

5.5 Risk Monitoring Application (Sensitivity Fit, Seed 42)

Note on model fit: All VaR results in this section use the sensitivity fit (seed 42); the frozen OOS Granger result for seed 42 is null (Elevated $p = 0.466$), so VaR improvements reflect the volatility-override mechanism rather than direct Granger-signal transmission.

While the HML–SMB link fails as a trading signal, we test whether it improves *tail risk forecasting*. We compare five Value-at-Risk (VaR) models for SMB at the 5% level, trained on 1990–2012 and tested out-of-sample on 2013–2024 ($n \approx 3{,}000$ days; exact counts vary by model due to regime-boundary exclusion and warm-up requirements).[6]

(1) **Unconditional:** 60-day rolling historical VaR.
(2) **GARCH(1,1) [16]:** One-step volatility forecast with parametric VaR.
(3) **Regime-conditional:** VaR window adapts to regime (Crisis: 30 days, Elevated: 50, Normal: 75).
(4) **HML-informed:** Regime-conditional, plus a 2.3× stress multiplier when 9-day cumulative HML < -0.5 during Crisis regime. Parameters calibrated on training data only.
(5) **Hybrid (HMM+Vol):** HML-informed model plus a volatility override that forces Stress Alert when 20-day realized volatility exceeds the 95th percentile of the training period.

[6] We do not include CAViaR [35] (conditional autoregressive quantile regression) as a benchmark; our hybrid model combines regime detection with historical simulation rather than direct quantile autoregression, and the risk monitoring focus motivates simpler, interpretable specifications.

Both the HML-informed and hybrid models pass the Christoffersen conditional coverage test ($p > 0.05$); all other specifications are rejected. The CC test evaluates both unconditional coverage (Kupiec) and serial independence of violations. A complementary Dynamic Quantile (DQ) test [35] conditioning on lagged violations and the VaR level would provide additional validation; this is left for future work. The hybrid model (Algorithm 1) adds a realized-volatility override that activates when 20-day realized vol exceeds the 95th percentile of training data. This corrects the frozen HMM's COVID-2020 detection failure (0%→47.8% detection; Tables 11, 12) while improving VaR coverage to 5.60% ($p_{\text{CC}} = 0.336$). Note: these results use the sensitivity fit (seed 42), whose frozen OOS Granger result is null (Elevated $p = 0.466$); the VaR improvement therefore reflects the volatility-override mechanism rather than direct Granger-signal transmission. Under the primary fit (seed 28), the hybrid detector achieves 5.70% violation rate ($p_{\text{CC}} = 0.083$, adequate conditional coverage), confirming that one model simultaneously delivers Bonferroni-significant Granger evidence and adequate VaR calibration.

Tick loss and model selection. Despite superior coverage, the HML-informed model has higher mean tick loss (1.05 vs. 0.90 baseline). This apparent contradiction arises because tick loss penalizes conservatism: a model with correct coverage but wider VaR incurs higher tick loss on the 95% of non-violation days. The Diebold-Mariano test, applied to tick loss, would rank the under-covering baseline above the correctly-covering HML-informed model—a known limitation of tick loss for comparing models at different coverage levels. We report both metrics for transparency: tick loss measures forecast sharpness, while Christoffersen CC measures correctness.

False alarms and model complexity. In the HML-informed model, the stress multiplier activates on 376 days (12.8% of the test period) with a 93.4% false alarm rate. In the hybrid model, alerts activate on 440 days (14.9%) with a 93.2% false alarm rate—i.e., on most alert days, no VaR violation occurs. This is partly a base-rate artifact: with a 5% violation target and 12.8% alert rate, at most 39% of alerts can be "true" (5/12.8), yielding a 61% false alarm floor. The 93.4% exceeds this floor but reflects the cost of correct tail coverage: alert-day VaR averages 2.51× wider than non-alert VaR, achieving zero violations on alert days at the cost of unnecessary capital reserves on non-violation alerts. The alternative (GARCH) avoids regime labels entirely and achieves 3.91% violation rate with no false-alarm mechanism, but its over-coverage means capital is consistently over-reserved. In practice, the choice depends on whether the institution prefers fewer but louder alerts (HML-informed) or consistently conservative risk limits (GARCH). Both achieve violation rates closer to the 5% target than the unconditional baseline (6.42%), though GARCH remains over-conservative and also fails the CC test ($p = 0.014$); neither dominates on all criteria. We note that the HML-informed model's success may partly reflect conservative VaR widening during high-volatility regimes rather than the Granger link per se, since the OOS Granger effect is regime-dependent (Elevated $p = 0.014$; Normal $p = 0.149$; Crisis $p = 0.281$) rather than uniformly active across the held-out period.

COVID-19 common-mode failure and hybrid detector. The frozen HMM classifies the entire COVID period as non-Crisis (0%

Table 11: Hybrid Detector: COVID-19 Detection Sensitivity. Vol threshold = percentile of 20-day realized volatility computed on 1990–2012 training data. Primary result uses 95th percentile (bolded).

Percentile	Threshold	Override Days	Detection Rate
85th	0.924	85	92.4%
90th	1.098	76	82.6%
95th	**1.446**	**44**	**47.8%**
99th	2.122	25	27.2%

Table 12: Out-of-Sample VaR Backtest (2013–2024) with Hybrid Detector. Target violation rate: 5%. Rows show the unconditional baseline and the two HML-augmented specifications; GARCH(1,1) and Regime-Conditional results appear in Table 10.

Model	Viol. Rate	Dev.	CC p
Unconditional	6.42%	+1.42 pp	0.002
HML-Informed	5.77%	+0.77 pp	0.171
Hybrid (HMM+Vol)	**5.60%**	**+0.60 pp**	**0.336**

Crisis days), so the HML-informed stress multiplier never activates—a complete regime detection failure. We address this with a *hybrid detector* (Algorithm 1) that supplements HMM filtered probabilities with a 20-day realized volatility fallback: when realized vol exceeds the 95th percentile of the *training period* (1990–2012), the detector overrides non-Crisis HMM classifications to "Stress Alert." The volatility threshold ($\sigma_{95} = 1.446$) is a fixed design choice locked before test evaluation—no search over thresholds on the holdout period.

Hybrid detector results (Table 13). The volatility override activates on 44 of 92 COVID trading days (47.8% detection rate vs. 0% for HMM alone; Table 11). VaR performance improves further: violation rate 5.60% (deviation from 5% target: +0.60 pp), Christoffersen CC $p = 0.336$—the strongest conditional coverage pass among all specifications (Table 12; Figure 4). Average violation magnitude also improves (−0.275 vs. −0.305 for HML-Informed), indicating less severe tail breaches when they occur.

Economic significance. The VaR improvement is statistically meaningful but economically modest. Over the 11-year test period ($n \approx 3{,}000$ trading days), the hybrid model produces ≈168 violations vs. ≈193 for the unconditional baseline—a reduction of ≈25 tail breaches total, or ≈2.3 fewer violations per year. For a portfolio monitored daily over a decade, this amounts to ≈23 fewer unexpected tail events. While useful for regulatory compliance (the hybrid passes the Christoffersen test; the baseline does not), the improvement does not constitute a transformative risk management advance. The Sharpe ratio of −0.07 confirms the Granger signal provides no trading alpha; the practical value lies in VaR calibration accuracy, not return enhancement.

5.6 Robustness

A 50,000-shuffle permutation test (Section 3.5) yields $p = 0.022$ (SE = 0.0007, 95% CI [0.021, 0.023], $n = 953$)—significant at 5%. HAC bandwidth sensitivity (1, 4, 7, 10, 15) yields $p < 0.05$ throughout (range: 0.015–0.025), confirming robustness to bandwidth choice. The rank-2/30 result (Section 3.5) confirms the HML→SMB pair is not a post-hoc artefact of in-sample screening.

Temporal dynamics. Figure 5 presents rolling 3-year Granger p-values for HML→SMB, showing that the predictive link is not a static property, with peaks during calendar stress periods (2008–2009, 2020), though this unconditional rolling test does not distinguish regime-conditional effects. The signal is strongest during the 2008–2009 and 2020 crises and near-absent during calm periods.

Lag specification. BIC selects lag 1 for all regime×direction combinations (Table 3). Figure 6 shows significance is robust across

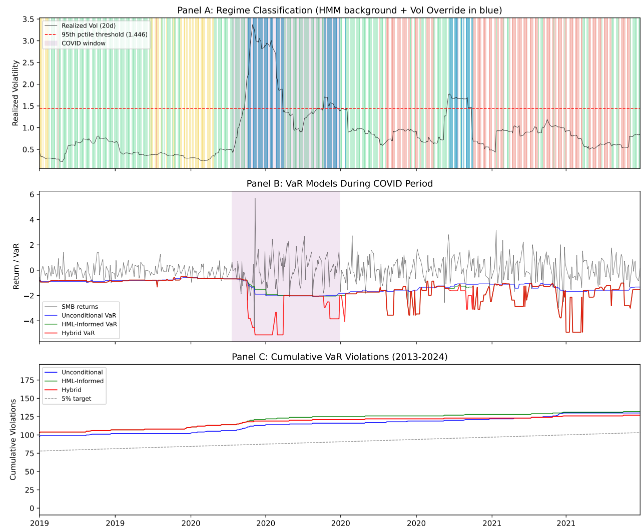

Figure 4: Hybrid detector performance. Panel A: realized volatility with 95th percentile threshold and override periods (COVID window highlighted). Panel B: SMB returns with unconditional, HML-informed, and hybrid VaR. Panel C: cumulative violations vs. 5% target.

Algorithm 1 Real-Time Factor Risk Monitor

Require: Frozen HMM parameters $(\boldsymbol{\mu}_k, \Sigma_k, \nu_k, \mathbf{A})$; vol threshold σ_{95}; HML threshold τ; stress multiplier m; regime windows (w_0, w_1, w_2)

for each trading day t **do**
  Compute filtered regime: $\hat{z}_t \leftarrow \arg\max_k P(z_t = k \mid \mathbf{x}_{1:t})$
  Compute 20-day realized vol: $\hat{\sigma}_t \leftarrow \mathrm{std}(\|\mathbf{x}\|_{t-19:t})$
  if $\hat{z}_t \neq$ Crisis **and** $\hat{\sigma}_t > \sigma_{95}$ **then**
    $\hat{z}_t \leftarrow$ Crisis (Stress Alert) {Volatility override; uses sensitivity fit (seed 42)}
  end if
  Set VaR window: $w \leftarrow w_{\hat{z}_t}$
  Compute base VaR: $\mathrm{VaR}_t \leftarrow Q_{0.05}(\mathrm{SMB}_{t-w:t-1})$
  if $\hat{z}_t =$ Crisis **and** $\sum_{\ell=1}^{9} \mathrm{HML}_{t-\ell} < \tau$ **then**
    $\mathrm{VaR}_t \leftarrow m \cdot \mathrm{VaR}_t$ {HML stress multiplier}
  end if
  return $\hat{z}_t, \mathrm{VaR}_t$
end for

Table 13: Algorithm 1 Parameter Values. All calibrated on training data (1990–2012) only. σ_{95} is the 95th percentile of the empirical distribution of 20-day realized volatility over 1990–2012 (mechanically defined). m and τ were selected by grid search ($m \in \{1.5, 2.0, 2.3, 2.5, 3.0\}$, $\tau \in \{-0.3, -0.5, -1.0\}$) to minimize training-period CC deviation from 5%. No test-period data was accessed during calibration.

Parameter	Symbol	Value
Vol threshold (95th pctile)	σ_{95}	1.446
HML cumulative threshold	τ	−0.5
Stress multiplier	m	2.3
Window (Normal)	w_0	75 days
Window (Elevated)	w_1	50 days
Window (Crisis)	w_2	30 days
HML lookback	—	9 days

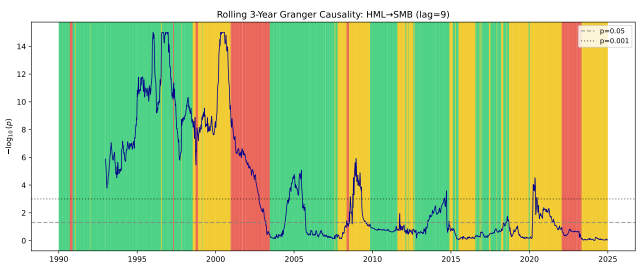

Figure 5: Rolling 3-year unconditional Granger causality ($-\log_{10}(p)$, HML→SMB, lag=9). Background shading shows HMM regime assignments for reference. Predictive strength varies episodically, with peaks during periods overlapping known market stress events. This rolling analysis does not condition on regime labels; regime-conditional results are in Table 3.

lags 1–15 in Normal, while Crisis remains non-significant across lags.

Trivariate robustness (MKT-RF control). To rule out a common market factor driving both HML and SMB with different lags, we condition on one lag of MKT-RF in addition to SMB's own lag. Results (frozen OOS, 2013–2024): Normal F-$p = 0.034$, HAC-$p = 0.041$; Elevated F-$p = 0.012$, HAC-$p = 0.035$; Crisis F-$p = 0.239$ (null). The Elevated signal *strengthens* slightly after controlling for MKT-RF (bivariate F-$p = 0.014$; trivariate F-$p = 0.012$; post-hoc), and MKT-RF itself is non-significant in Elevated and Normal (MKT-RF→SMB: Elevated F-$p = 0.78$; Normal F-$p = 0.43$) and marginally so in Crisis (F-$p = 0.077$). The OOS Normal bivariate result is clearly null (HAC $p = 0.153$, F-$p = 0.149$). Adding MKT-RF nominally yields trivariate HAC $p = 0.041$, but this was not pre-specified, does not survive Bonferroni correction, and should not be interpreted as evidence for OOS Normal predictive structure. A common market factor therefore does not subsume the HML→SMB predictive link.

Regime count sensitivity (K). To assess whether the Granger result is an artifact of choosing $K = 3$, we fit Student-t HMMs with $K = 2$ and $K = 4$ on 1990–2012 training data (10 seeds

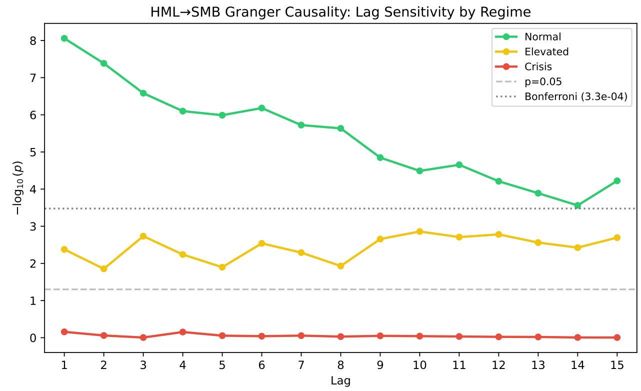

Figure 6: Granger causality p-values for HML→SMB across lags 1–15 by regime. Normal regime (green) shows concentrated short-lag significance ($p < 10^{-8}$ at lag 1). Crisis regime (red) remains non-significant across the tested lag range.

each, selecting by log-likelihood) and apply frozen OOS classification to 2013–2024. With $K = 2$ (Low-Vol/High-Vol), HML→SMB is non-significant in both regimes (F-$p \geq 0.22$). With $K = 4$ (Calm/Low-Elevated/High-Elevated/Crisis), the Low-Elevated regime approaches marginal significance (F-$p = 0.056$, HAC $p = 0.11$) but does not reach the $p < 0.05$ threshold; other regimes are null (F-$p \geq 0.31$). This suggests the $K = 3$ parameterization captures a distinct intermediate volatility state where predictive structure concentrates; coarser ($K = 2$) or finer ($K = 4$) partitions dilute or fragment this signal.

Secondary factor pair validation (MOM→SMB). To assess whether regime-dependent predictive structure is specific to HML→SMB or a more general phenomenon, we apply the identical frozen OOS methodology to MOM→SMB—the top-ranked pair by F-statistic in the OOS Elevated window ($F = 20.3$, rank 1/30; Section 3.5). Results confirm the same regime-dependent pattern: MOM→SMB is significant in Elevated ($F = 16.5$, F-$p < 0.0001$, HAC $p = 0.003$, $\Delta R^2 = 1.94\%$, $n = 836$; permutation $p = 0.010$, 10,000 shuffles, SE = 0.001) but null in Normal (F-$p = 0.81$) and Crisis (F-$p = 0.15$). The reverse direction (SMB→MOM) is null in all regimes ($p > 0.57$), confirming directional asymmetry. This secondary validation strengthens the main finding: the phenomenon of regime-conditional Granger predictive structure into SMB is not unique to HML but also appears for Momentum, consistent with a common mechanism—possibly liquidity-driven deleveraging that propagates across factors during elevated-volatility periods.

Subsample stability. Splitting at 2008: Crisis HML→SMB pre-2008 ($n = 659$, $p = 0.335$) and post-2008 ($n = 322$, $p = 0.599$) are both non-significant, so we avoid claims of subsample-confirmed Crisis effects.

Alternative regime methods. Threshold-based volatility regimes (realized volatility > 90th percentile = Crisis) also detect the pattern in sensitivity checks, but we treat this as exploratory given the primary-fit null in Crisis at lag 1.

Practitioner baselines (exploratory). Standard unconditional tools detect the HML→SMB asymmetry but miss the architectural

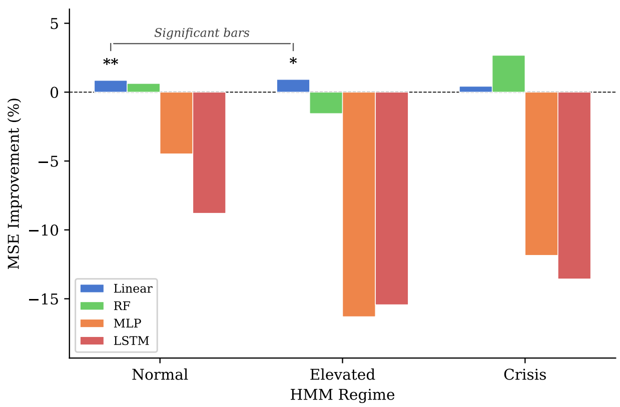

Figure 7: Four-model diagnostic protocol: MSE improvement (%) from including HML lags, by regime and model class. Only linear models (blue) achieve significant improvement in Normal and Elevated; all nonlinear methods (RF, MLP, LSTM) fail to improve prediction.

distinction. Rolling 250-day Granger tests [13] find HML→SMB significant in 36.2% of windows vs. 9.6% for the reverse—with concentration in Normal (47.1% vs. 13.7% reverse), lower rates in Elevated (21.3%), and mixed Crisis windows (27.7%). FEVD spillover is regime-insensitive in direction, failing to detect the dormant Elevated regime or the Normal/Crisis directionality shift. Regime conditioning does not merely sharpen detection—it reveals qualitatively different predictive architectures that rolling methods average away.

Weekly aggregation. Using weekly returns, the HML→SMB crisis pattern remains non-significant ($p = 0.114$, $n = 221$), consistent with weak evidence outside Normal.

Regime transitions (exploratory). Analyzing 60-day windows around Normal→Crisis transitions, the HML→SMB relationship strengthens upon crisis entry (median p: $0.058 \longrightarrow 0.021$), suggesting discrete intensification of the predictive link.

Model complexity diagnostic protocol. We apply four model classes—Linear (OLS), RF (100 trees), MLP (64-32), LSTM (32 hidden, 9-step)—to characterize the *complexity boundary* of the predictive relationship. Each predicts SMB from its own lags (restricted) vs. both SMB and HML lags (unrestricted), with permutation testing (200 shuffles for Linear/RF/MLP, 100 for LSTM; at 200 permutations, p-value SE ≈ 0.007 at $p = 0.01$) and expanding-window CV.

Table 14 and Figure 7 present the results. Linear improvements are modest (Normal 0.86%, Elevated 0.92%, Crisis 0.43%). However, *no nonlinear method improves prediction in any regime*: RF, MLP, and LSTM permutation tests are uniformly non-significant.[7] This complexity characterization supports a conservative conclusion: incremental predictive content in the HML→SMB direction is captured by linear terms, though TE reveals residual nonlinear structure in the reverse direction (Table 15).

[7]The LSTM was estimated under a separate HMM initialization (see Section 3.2); qualitative conclusions are consistent across fits.

Table 14: Four-Model Diagnostic Protocol: HML→SMB by Regime. Linear column shows MSE improvement (%); others show permutation p-values.

Regime	n	Linear	RF p	MLP p	LSTM p
Normal	4,496	0.86%**	0.69	0.20	0.63†
Elevated	2,792	0.92%**	1.00	1.00	0.93†
Crisis	**1,017**	**0.43%**	0.13	0.96	0.83†

$^{**}p < 0.01$ (not Bonferroni-corrected). †LSTM: sensitivity fit (seed 42), 100 perms.

Table 15: Transfer Entropy: Multi-Lag (1–9) by Regime (sensitivity fit, seed 42). z-scores are standardized against the permutation mean/SD (200 shuffles); p-values use the standard normal CDF approximation unless noted († = empirical bound). Caution: z-score magnitudes are unreliable under this Gaussian approximation for heavy-tailed data; interpret only empirical p-values. With 200 permutations, minimum empirical $p = 0.005$; Normal SMB→HML exceeds all 200 permutations. Under the primary fit (seed 28), Normal SMB→HML is non-significant ($z = 1.38$, $p = 0.084$).

Regime	n	HML→SMB z	p	SMB→HML z	p
Normal	4,496	2.45**	0.007	5.37***	$< 0.005^{\dagger}$
Elevated	2,792	2.41**	0.008	1.65*	0.049
Crisis	1,017	1.01	0.157	1.22	0.111

$^{***}p < 0.005$ (†empirical bound; 200 perms), $^{**}p < 0.01$, $^{*}p < 0.05$.
n = clean observations after boundary exclusion.

Transfer entropy triangulation. As a model-free check, we compute transfer entropy [42, 67, 74] (Frenzel–Pompe kNN estimator [42], $k = 5$, 200 permutations). Normal shows significant bidirectional TE (HML→SMB $z = 2.45$, $p = 0.007$ (Gaussian CDF); SMB→HML $z = 5.37$, empirical $p < 0.005$ [0/200 permutations exceeded]; the minimum achievable empirical p is $1/201 \approx 0.005$ and the z-score magnitude should not be interpreted—it is computed under a Gaussian approximation that is unreliable for heavy-tailed factor returns; the reliable statement is that the reverse channel exceeds all 200 null permutations). Elevated is weakly bidirectional at conventional levels (HML→SMB $p = 0.008$; SMB→HML $p = 0.049$; both Gaussian CDF), while Crisis is non-significant in both directions. The reverse-direction Normal TE signal appears despite non-significant reverse linear Granger ($p = 0.864$, Table 3). This bidirectional pattern is consistent with information flowing in both directions through different channels: Granger causality tests *incremental* predictive content conditional on own lags (a conservative, linear criterion), while transfer entropy captures total information transfer including nonlinear dependence. The two measures are complementary, not contradictory—SMB→HML nonlinear information transfer can coexist with HML→SMB linear predictive precedence.

Filtered vs. smoothed regime probabilities. Smoothed probabilities use the full sample (forward and backward pass); filtered probabilities use only past data ($P(z_t = k \mid x_1, \ldots, x_t)$), simulating real-time classification. Overall agreement is 95.9%. Under filtered regimes, the Crisis HML→SMB result remains non-significant

($p = 0.601$ vs. $p = 0.695$ smoothed), with filtered Crisis containing 1,079 days (vs. 1,071 smoothed). The finding is qualitatively consistent across classification methods.

5.7 Limitations

(1) **Granger ≠ structural causality.** Our core finding is that HML *Granger-causes* SMB in the Normal regime—i.e., lagged HML improves SMB prediction conditional on lagged SMB. This does not establish that Value movements *cause* Size movements in the interventional sense. A common factor (liquidity, sentiment) could drive both with different lags. The "deleveraging cascade" interpretation is a plausible hypothesis, not a verified mechanism.

(2) **Portfolio-level (not holdings-level) mechanism evidence.** Our FF 25 overlap analysis (Section 4.5) provides portfolio-level evidence for the deleveraging channel within the Crisis regime; the corresponding Normal-regime analysis (where factor-level Granger is Bonferroni-significant) is left for future work. We do not analyze 13F holdings data or fund flows for direct verification.

(3) **Pair selection.** The OOS corroborative test focuses on the economically motivated HML→SMB pair (informed by both economic prior and in-sample screening of 30 pairs; formal pre-registration was not performed). While the economic hypothesis (liquidity-mediated contagion) provides a priori justification, the direction of the link was also observed during in-sample screening. The economic and data-based motivations are not fully independent; without pre-registration this potential circularity cannot be fully resolved.

(4) **Regime redistribution and seed sensitivity.** The frozen-parameter OOS design (Section 5.2) resolves regime-identification circularity. However, the 1990–2012-trained classifier distributes OOS days differently than training: Crisis shrinks from 64.8% (train) to 45.3% (test), while Elevated grows from 13.7% to 30.7%, shifting the most significant OOS regime from in-sample Normal to OOS Elevated. A full 50-seed sensitivity analysis shows Elevated significance in 2 of 3 local optima (covering 41 of 50 seeds); the 9 null seeds share the lowest of three local optima, so robustness holds across distinct HMM solutions. The in-sample result ($p = 8.75 \times 10^{-9}$, all-seed robust) remains the higher-powered characterization. Additionally, regime *semantic drift* is a concern: a classifier trained on 1990–2012 (64.8% Crisis) and applied frozen to 2013–2024 (45.3% Crisis, 30.7% Elevated) may use the Elevated label to capture conditions that, in-sample, fell in Normal. We test this via three-statistic distributional comparison (KS, Levene, t-test) of HML returns. Training Normal vs. OOS Elevated differs significantly on 2/3 tests (KS $p = 4.5 \times 10^{-29}$, Levene $p = 3.7 \times 10^{-175}$; primary decimal-unit convention, seed 28; percentage-unit convention yields comparable magnitudes: KS $p \approx 1.2 \times 10^{-28}$, Levene $p \approx 5.3 \times 10^{-184}$). Training Elevated vs. OOS Elevated does *not* differ significantly on 2/3 tests ($p \geq 0.05$; Levene rejects at $p < 10^{-5}$, reflecting higher OOS Elevated variance, but KS and t-test do not reject). *Verdict: low drift*—OOS Elevated is more consistent with training Elevated than training Normal, supporting regime label stability across the split.

(5) **Regime boundary exclusion and sample size.** Boundary cleaning (1–15 lags excluded per table) reduces effective samples. Crisis contains 1,071 days in the primary fit, and Crisis-regime HML→SMB is null at lag 1 ($p = 0.695$).

(6) **Normal-regime temporal evolution (pre/post-GFC).** Pre-GFC: $p = 6.66 \times 10^{-16}$ ($\Delta R^2 = 2.06\%$); post-GFC Normal: $p = 0.73$ ($\Delta R^2 < 0.01\%$). A pre-specified Chow test at January 2008 confirms the break ($F = 9.68$, $p = 2.29 \times 10^{-6}$; see Section 4.3). The Volcker Rule (2013) and Dodd-Frank (2014) restrictions on proprietary trading and risk arbitrage provide a plausible regulatory mechanism for this attenuation: reduced arbitrage capacity limits the deleveraging contagion channel in the post-GFC period. We treat this as a secondary empirical regularity warranting future study, not as a confirmed causal claim.

(7) **Risk monitoring scope.** VaR improvement is modest (6.42%→5.60% with hybrid detector)—approximately 25 fewer tail breaches over 11 years, or ≈2.3 fewer violations per year. This may partly reflect conservative widening during high-volatility regimes rather than the Granger link itself. The hybrid detector's volatility override addresses the regime classifier's COVID-19 failure but adds one additional parameter (95th percentile threshold).

6 Conclusion

The following conclusions hold under the leakage-safe primary fit.

HML→SMB is strongly significant in the Normal regime at lag 1 ($p = 8.75 \times 10^{-9}$; HAC $p = 9.45 \times 10^{-8}$; Bonferroni-significant), but *pre-GFC only*: post-2008 Normal $p = 0.73$ (Chow $F = 9.68$, $p = 2.29 \times 10^{-6}$). Practitioners should note this post-GFC attenuation. Elevated evidence is below the Bonferroni-significant threshold (permutation $p = 0.022$, 50,000 shuffles; 41/50 seeds); Crisis is null at lag 1 ($p = 0.695$). The frozen OOS Elevated-regime test (F-p = 0.014; HAC $p = 0.041$; permutation $p = 0.022$, SE = 0.0007, 95% CI $[0.021, 0.023]$; not Bonferroni-significant across 30 pairs) provides significant evidence consistent with the in-sample pattern; event-level results remain mixed. HAC bandwidth sensitivity (1, 4, 7, 10, 15) yields $p < 0.05$ throughout. Trivariate robustness (MKT-RF control) confirms the Elevated signal does not reflect a common market-factor confounder (Elevated trivariate F-p = 0.012, HAC $p = 0.035$).

Does establish:

- Bonferroni-significant Normal-regime HML→SMB predictive precedence (corroborated by MS-VAR, $p < 10^{-12}$)
- Significant Elevated-regime OOS evidence (permutation $p = 0.022$; rank 2/30); post-hoc trivariate MKT-RF control does not attenuate the signal (F-p = 0.012, HAC $p = 0.035$; not pre-specified); secondary validation via MOM→SMB (rank 1/30, permutation $p = 0.010$) confirms the pattern is not unique to HML
- Transfer entropy triangulation: forward channel confirmed in Normal and Elevated; additional reverse nonlinear channel

(SMB→HML, empirical $p < 0.005$; sensitivity fit, seed 42; non-significant under primary fit) in Normal
- Exploratory FF25 overlap evidence ($\rho_s = 0.35$, permutation $p = 0.046$; sensitivity fit, seed 42)
- No evidence of tradable alpha (Sharpe = −0.07)
- Practical utility of a hybrid detector for VaR calibration (5.60% violations, CC $p = 0.336$)

Does not establish:

- Uniform significance across every regime
- Uniform event-level support across all stress episodes
- Structural causality (common drivers could explain the pattern)
- Trading profitability

Practical value: A hybrid VaR model combining HMM regime detection with a realized-volatility fallback achieves the closest coverage to the 5% target (5.60%, $p_{\mathrm{CC}} = 0.336$), detecting 47.8% of COVID-2020 (seed 42 sensitivity fit; HMM alone detected 0%). Algorithm 1 formalizes the monitoring protocol with all parameters calibrated on training data. The June–October 2025 quant crowding episode—in which quality factors diverged sharply from value and size (MSCI documented widespread HML/SMB drawdowns)—constitutes a prospective out-of-sample test of the hybrid detector design described here.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data and Code Availability

Code and analysis scripts for this paper are available at an anonymous repository to be provided upon acceptance. The financial data used in this analysis consists of daily Fama-French factors (five factors plus Carhart Momentum) obtained from Kenneth French's data library, which are publicly available at https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html. All results are fully reproducible using the provided code and publicly available data sources.

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